# The Impact of USDM

## From Documents to Digital Workflows

Dave Iberson-Hurst, March 17, 2026
ACDM Berlin

LinkedIn

# Dave Iberson-Hurst

*Dave has over 40 years' experience across several industries with the last 25+ years spent in the pharmaceutical industry combining his technology and software development experience with clinical trial process and data standards knowledge.*

*During this time, he has served as the CDISC CTO, worked on, and led, several CDISC teams, presented in many forums in Europe, the US, and elsewhere across the globe. He has worked closely with the FDA, EMA, HL7, ISO, and other standards organizations and was was a member of CDISC's Blue Ribbon commission.*

*From January 2022 to December 2025, he served (on contract to CDISC) as the CDISC Product Owner and Technical Expert for the Digital Data Flow / Unified Study Definitions Model project.*

*He is a partner at data4knoweldge in Copenhagen and is focused on getting greater value and utility from clinical trial data.*

# Agenda

- Introduction
- DDF, USDM and M11
- Industry Vision and Impact
- Use Cases
- Summary
- Links

# Introduction

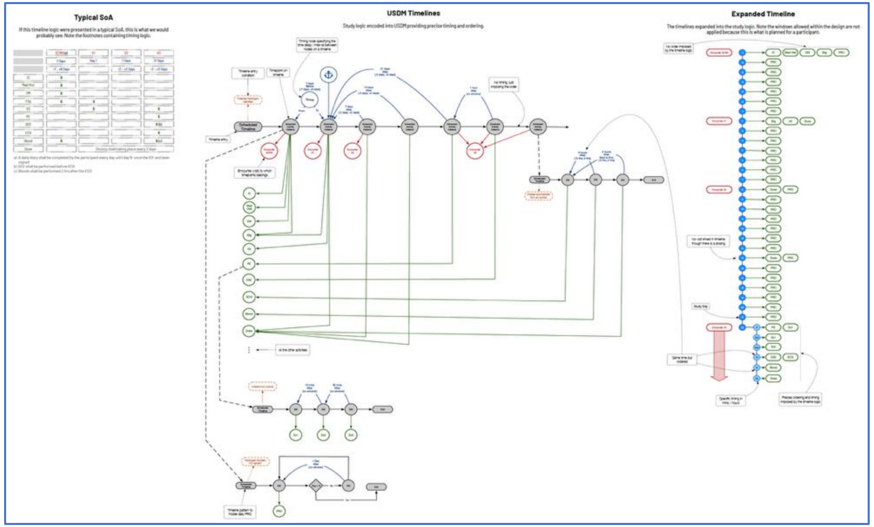
# Clinical Trial Complexity

## Typical SoA

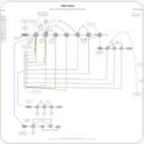
If this timeline logic were presented in a typical SoA, this is what we would probably see. Note the footnotes containing timing logic.

| | SCRN (a) | V1 | V2 | V3 |
|---|---|---|---|---|
| | 7 Days | Day 1 | 7 Days | 21 Days |
| | -7 .. +6 Days | | -2 .. +2 Days | -2 .. +2 Days |
| IC | X | | | |
| Med Hist | X | | | |
| DM | X | | | |
| Elig | X | X | | |
| VS | | X | | X |
| PE | | | | X |
| QS2 | | | | X (b) |
| ECG | | | | X |
| Blood | X | | | X (c) |
| Dose | | Dosing shall taking place every 7 days | | |

a) A daily diary shall be completed by the participant every day until day N once the ICF and been signed
b) QS2 shall be performed before ECG
c) Bloods shall be performed 2 hrs after the ECG



# Clinical Trial Complexity



Using this diagram provide a single web page which displays the study expectations from the perspective of a subject wishing to enroll in the study

PARTICIPANT INFORMATION

## Your Guide to the Study

This page explains what will happen at each stage of the study, what we'll ask you to do, and roughly how long each visit will take. We want you to feel informed and prepared.

| | | |
|---|---|---|
| **4**<br>Clinic Visits | **~22**<br>Days in Study | **Daily**<br>Short Diary Entry |
| **Every 7d**<br>Dosing Schedule | | |

## Your Visit-by-Visit Journey

Here's what happens at each visit. Scheduling windows are included so you know how flexible appointment timing can be.

### Screening Visit
Day -7 *Window: up to 6 days before Visit 1*

This is your first visit to the clinic. We'll go through the study together, answer your questions, and check whether you're eligible to take part.

- **Informed Consent** — We'll explain the study and you can decide if you'd like to participate
- **Medical History** — We'll ask about your health background
- **Demographics** — Basic information like age and background
- **Eligibility Check** — We'll confirm you meet the study criteria

### Visit 1 — Enrolment
Day 1 *Window: ±2 days*

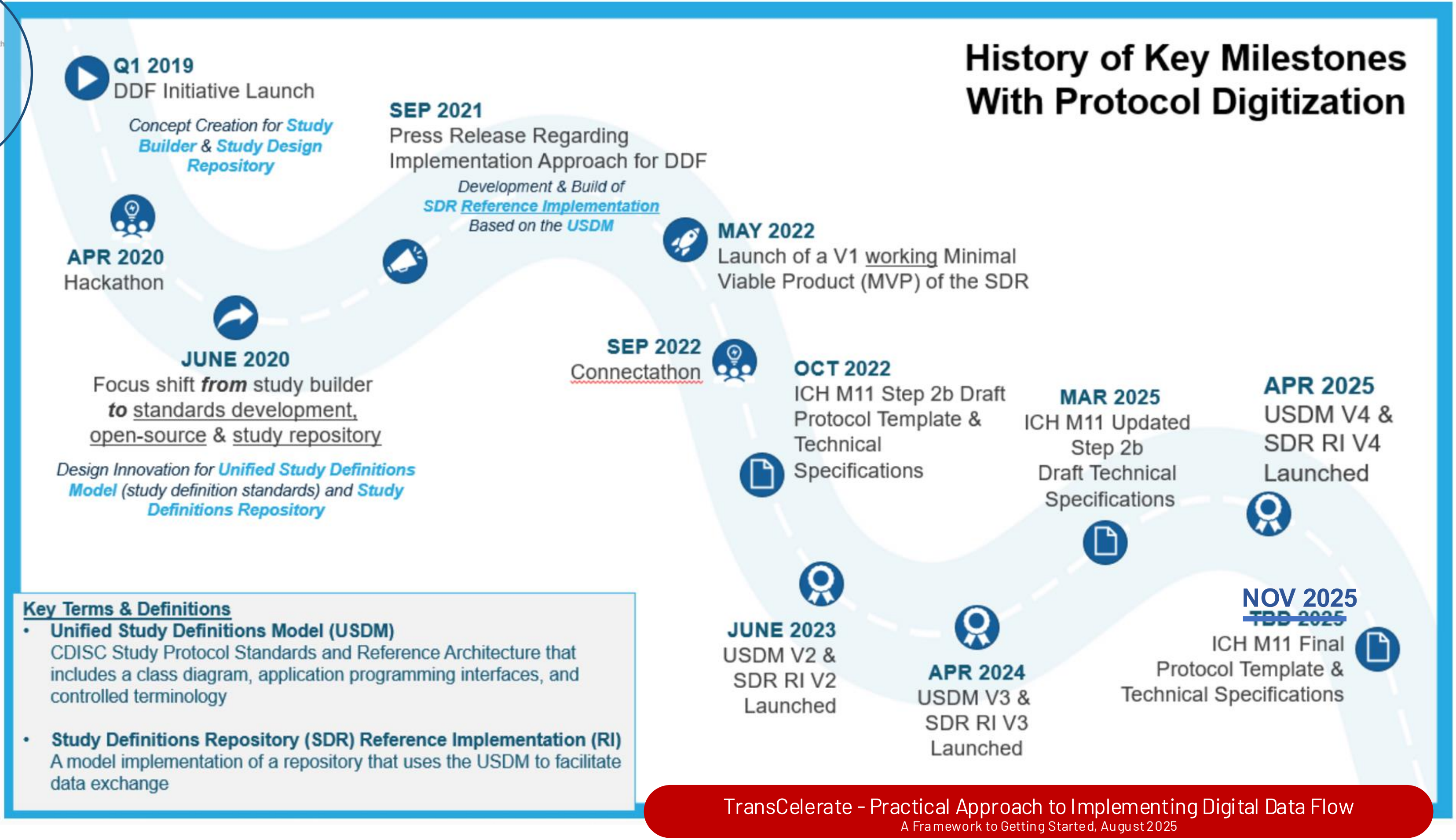
If you're eligible, this is when the study officially begins. We'll do some initial assessments and you'll start your daily diary.

- **Eligibility Confirmation** — Final check before enrolment
- **Vital Signs** — Blood pressure, heart rate, temperature

**Daily diary starts now:** Once you're enrolled, you'll complete a short daily questionnaire (PRO) every day until the end of the study.

# DDF, ICH M11 and USDM

M11
June 2018



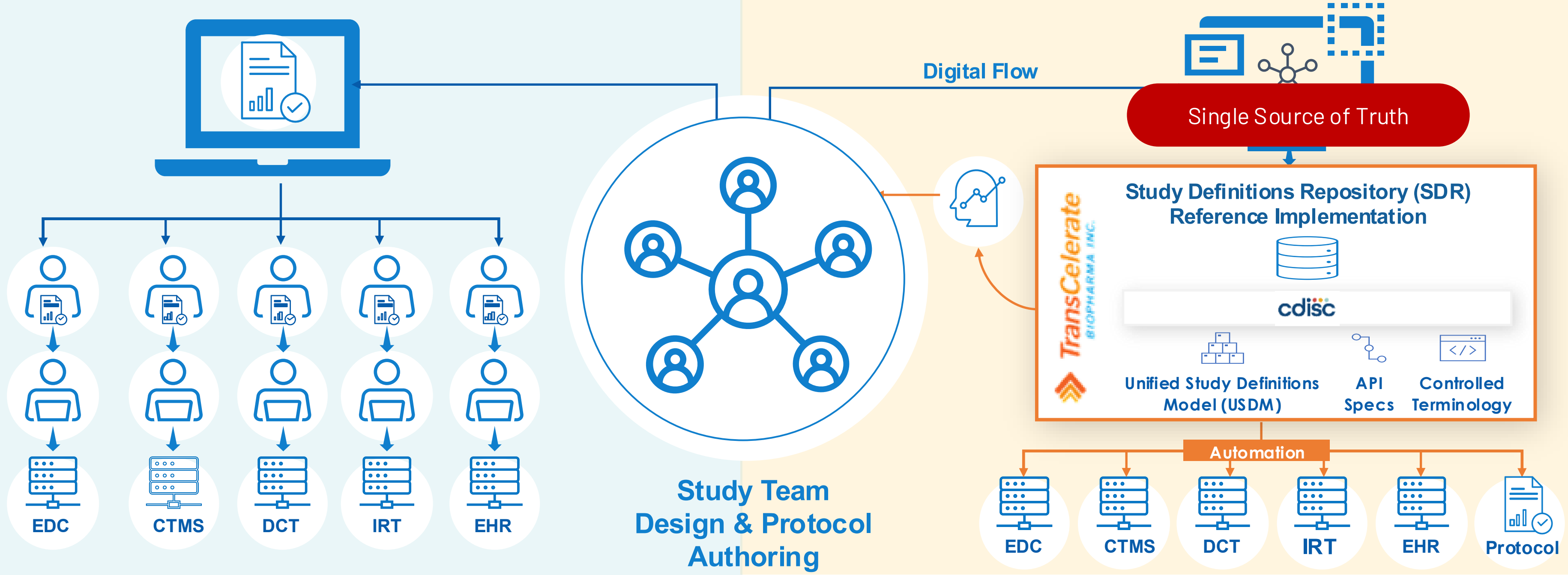
https://www.transceleratebiopharmainc.com/assets/digital-data-flow-solutions/

# TransCelerate Digital Data Flow (DDF) Ambition

*Write Once, Read Many*

**TODAY:** Document-based paradigm for protocol creation, interpretation, and transcription into consuming systems

**TOMORROW:** Digital paradigm for protocol creation, with fully automated data flow and interoperability between systems






TransCelerate
BIOPHARMA INC.

# ICH M11 Adopted

ICH
harmonisation for better health

**Final Guidelines adopted by the Assembly *(Step 4 of the ICH process)***

- The M11 Guideline on *Clinical Electronic Structured Harmonized Protocol (CeSHarP), M11 Clinical Implementation Template and M11 Technical Specification* was adopted and has now entered the implementation phase. The Guideline describes general protocol design principles and approach used to develop the separate associated documents, the ICH M11 Clinical Electronic Structured Harmonised Protocol Template and the Technical Specification that are acceptable to all regulatory authorities of the ICH regions. The Template presents the format and structure of the protocol, including table of contents, common headers, and instructions for content. The Technical Specification presents the data elements and technical attributes (e.g., definition, conformance, cardinality) that enable the interoperable electronic exchange of protocol content.

**M11 Versions**

- 2022-10-14 (First Review)
- 2025-03-14 (Second Review)
- 2025-11-16 (Adopted)

**M11 Clinical electronic Structured Harmonised Protocol (CeSHarP)**

**M11 EWG** **Clinical electronic Structured Harmonised Protocol (CeSHarP)**

This topic was endorsed by the ICH Management Committee in November 2018. This new guideline is proposed to provide comprehensive clinical protocol organisation with standardised content, with:

- A Template which presents the format

**Guideline**

- M11 Guideline
- M11 Technical Specification
- M11 Template

# ICH M11 Documents

ICH
harmonisation for better health

INTERNATIONAL COUNCIL FOR HARMONISATION OF TECHNICAL REQUIREMENTS FOR PHARMACEUTICALS FOR HUMAN USE

ICH HARMONISED GUIDELINE

**CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL**

**(CESHARP)**

**M11 TEMPLATE**

*Final version*

*Adopted on 19 November 2025*

ICH M11 Template

ICH
harmonisation for better health

INTERNATIONAL COUNCIL FOR HARMONISATION OF TECHNICAL REQUIREMENTS FOR PHARMACEUTICALS FOR HUMAN USE

ICH HARMONISED GUIDELINE

**CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL**

(CeSHarP)

**M11 TECHNICAL SPECIFICATION**

Final version

*Adopted on 19 November 2025*

ICH M11 Technical

# Phase Example

**Investigational Product Name(s):**

<Enter Nonproprietary Name(s)>

<Enter Proprietary Name(s)>

Omit nonproprietary name field if a nonproprietary name has not yet been assigned. Omit proprietary name field if not yet established.

**Trial Phase:** [Trial Phase]

For trials combining investigational drugs or vaccines with devices, classify according to the phase of drug development.

**Short Title:** <Enter Trial Short Title>

Short title should convey in plain language what the trial is about and shou
Plain Language"
suitable for use
committee subr

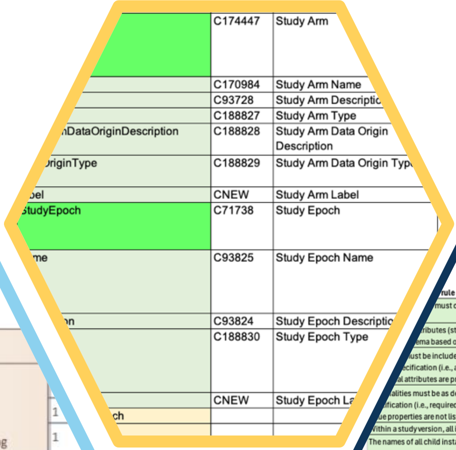
| | |
|---|---|
| **Term (Variable)** | [Trial Phase] |
| **Data Type** | Valid Value |
| **Data (D), Value (V) or Heading (H)** | V |
| **Definition** | C48281 |
| **User Guidance** | For trials combining investigational drugs or vaccines with devices, classify according to the phase of drug development. |
| **Conformance** | Required |
| **Cardinality** | One to one; Sponsor Protocol Identifier |
| **Relationship content from ToC representing the protocol hierarchy** | Title Page |
| **Value** | Code List C217045: ICH OID 2.16.840.1.113883.3.989.2.3.1.18 Early Phase 1 (C54721); Phase 1 (C15600); Phase 1/Phase 2 (C15693); Phase 1/Phase 2/Phase 3 (C198366); Phase 1/Phase 3 (C198367); Phase 2 (C15601); Phase 2/Phase 3 (C15694); Phase 2/Phase 3/Phase 4 (C217024); Phase 3 (C15602); Phase 3/Phase 4 (C217025); Phase 4 (C15603) |
| **Business rules** | **Value Allowed**: Yes<br>**Relationship**: Heading; Sponsor Protocol Identifier<br>**Concept:** C48281 ICH OID 2.16.840.1.113883.3.989.2.3.3.16 |
| **Repeating and/or Reuse Rules** | No |

| | |
|---|---|
| **Term (Variable)** | Trial Phase: |
| **Data Type** | Text |
| **Data (D), Value (V) or Heading (H)** | H |
| **Definition** | Heading |
| **User Guidance** | N/A |
| **Conformance** | Required |
| **Cardinality** | One to one; One to Sponsor Protocol Identifier |
| **Relationship content from ToC representing the protocol hierarchy** | Title Page |
| **Value** | Trial Phase: |
| **Business rules** | **Value Allowed**: No<br>**Relationship**: Table row heading; Sponsor Protocol Identifier<br>**Concept:** Heading |
| **Repeating and/or Reuse Rules** | No |

# The USDM Standard

## CDISC Controlled Terminology

Provides further semantics, complementing the UML model. Includes the definition of classes, attributes, and value sets

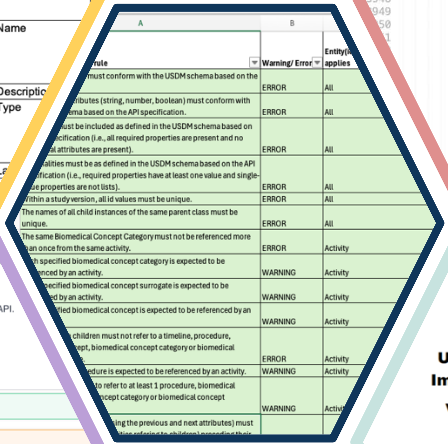



## Examples

Example protocols implemented in the USDM with associated JSON files and visualisations

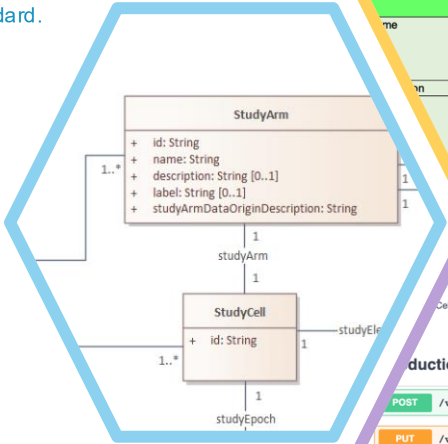

## CORE Rules

Specification of the rules that define USDM compliance

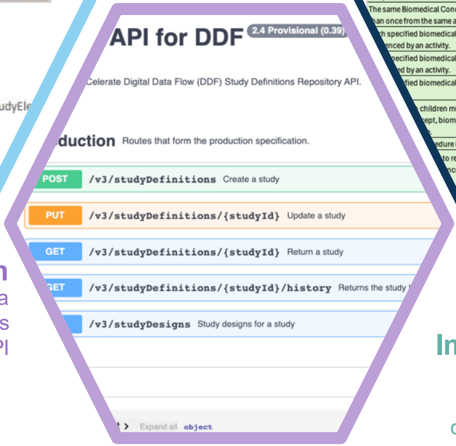



## Logical Model

The UML logical model (a class diagram) that provides the basis for the USDM standard.

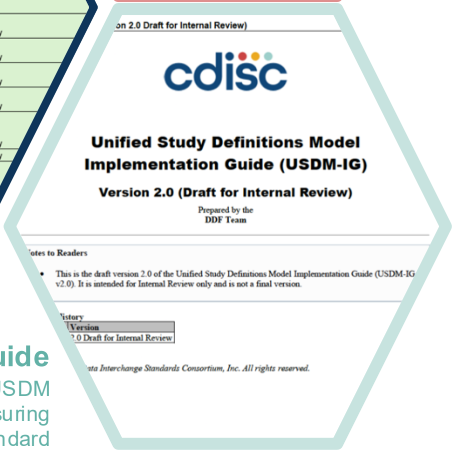



CDISC GitHub




## API Specification

Provides the means to exchange a single study between machines using a JSON API

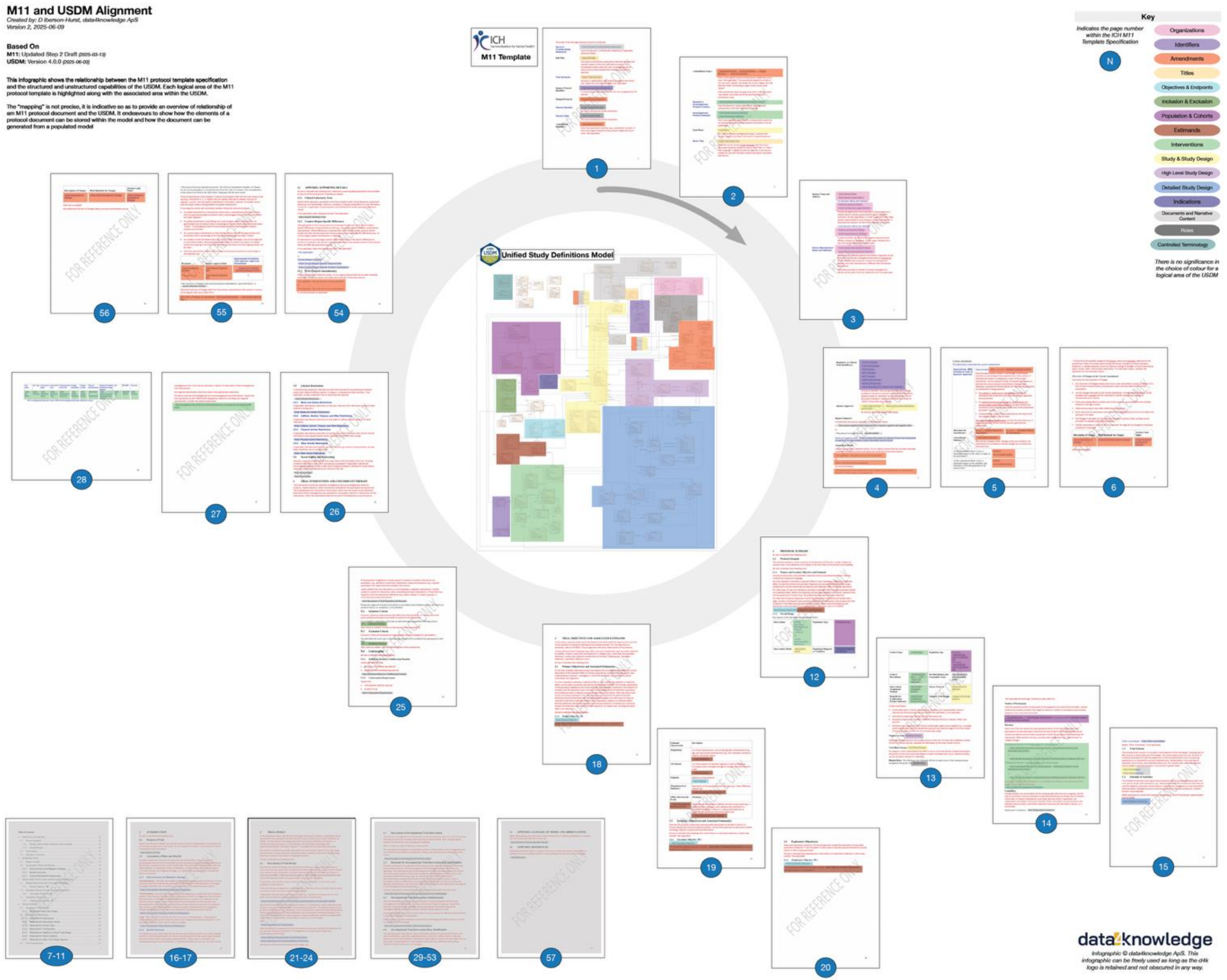



## Implementation Guide

Guidance on using the USDM model and ensuring conformance with the standard

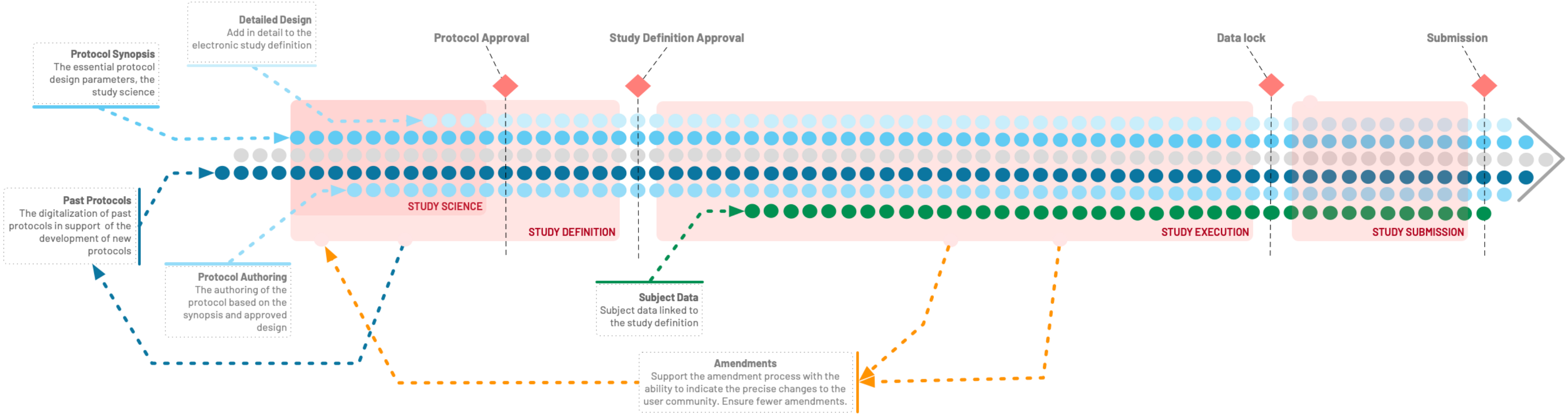
The graphic provides a visual guide to how the content defined by the ICH M11 Teamplate speciciation is handled within USDM

Infographic



# Industry Vision and Impact

# From Silos to Continuum

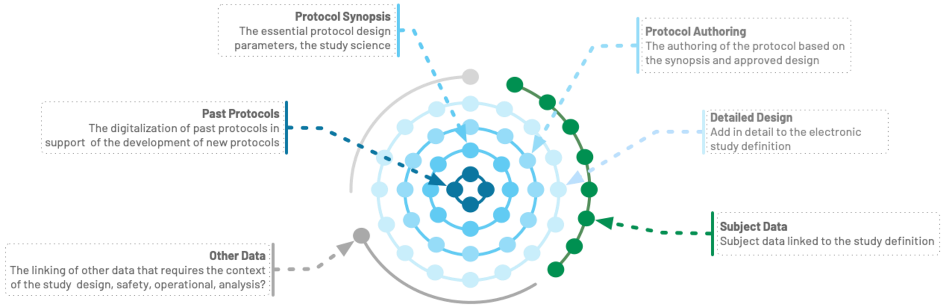



**Consistency**

USDM's standardized framework helps eliminates data silos that have plagued clinical research by providing a consistent single source of truth to diverse platforms.

**Capacity**

The existence of a single, electronic definition eliminates the need for manual transcription from PDF protocols into multiple systems, reducing human error and accelerating execution times. Saving just a few hours on each activity quickly accumulates to substantial savings across the entire study life-cycle increasing an organization's capacity.




**Context**

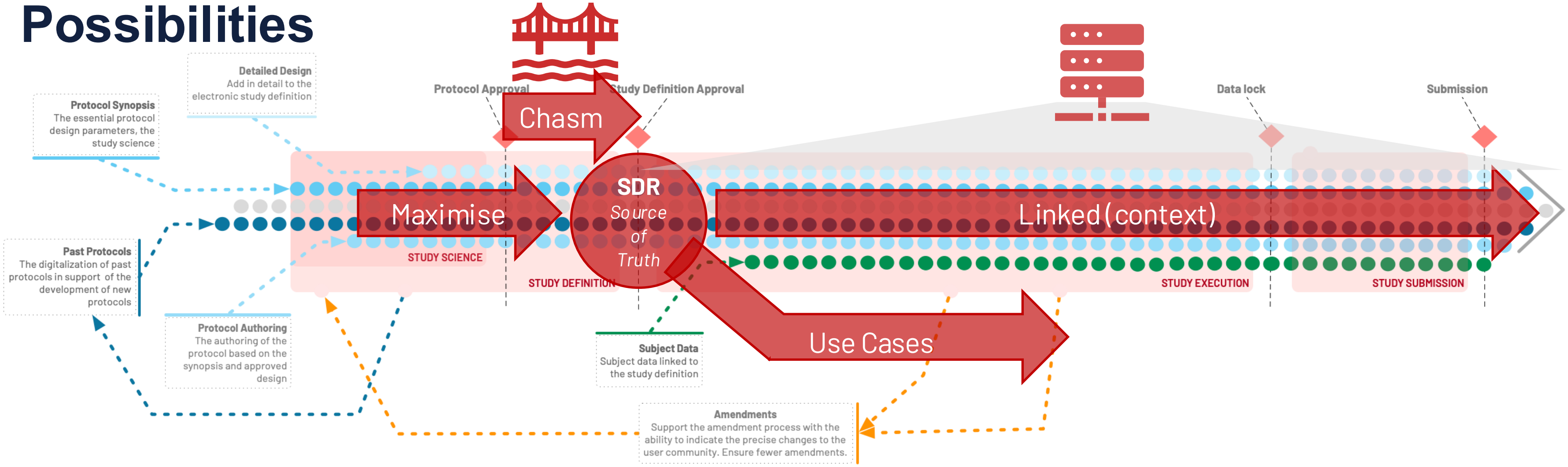
USDM & BCs serve as a catalyst for AI technologies by providing data with the full trial design context resulting in improved data quality, and enhanced cross-study consistency. It addresses the critical limitation that "AI doesn't solve our context problems; it amplifies them" by bringing essential context to clinical trial data, empowering AI to operate more effectively.

**Compliance**

A shared data model fosters better communication and alignment not only with among sponsors, CROs, and vendors but also with regulators. The combination of USDM & ICH M11 will facilitate dialogue and regulatory compliance across the globe.

Draft 1, 15th December 2025. Prepared by Dave Iberson-Hurst

# Possibilities

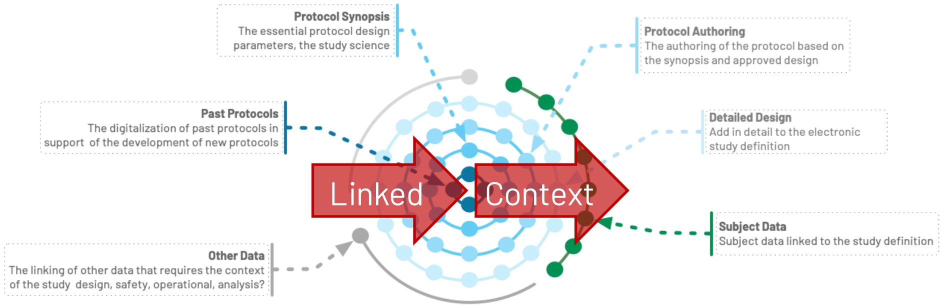



**Consistency**

USDM's standardized framework helps eliminates data silos that have plagued clinical research by providing a consistent single source of truth to diverse platforms.

**Capacity**

The existence of a single, electronic definition eliminates the need for manual transcription from PDF protocols into multiple systems, reducing human error and accelerating execution times. Saving just a few hours on each activity quickly accumulates to substantial savings across the entire study life-cycle increasing an organization's capacity.




**Context**

USDM & BCs serve as a catalyst for AI technologies by providing data with the full trial design context resulting in improved data quality, and enhanced cross-study consistency. It addresses the critical limitation that "AI doesn't solve our context problems; it amplifies them" by bringing essential context to clinical trial data, empowering AI to operate more effectively.

**Compliance**

A shared data model fosters better communication and alignment not only with among sponsors, CROs, and vendors but also with regulators. The combination of USDM & ICH M11 will facilitate dialogue and regulatory compliance across the globe.

Draft 1, 15th December 2025. Prepared by Dave Iberson-Hurst

# Use Cases

# USDM in Action
# Use Cases Supporting the DDF Vision

## Use Cases

### Retrospective

### Protocol 'Store'

**Past Protocols**
The storage of past sponsor protocols in to support a variety of use cases such as:

- Therapeutic area specific content
- Standard inclusion and exclusion criteria.
- Libraries of objectives, endpoints and estimands.
- Asses the past to prevent protocol amendments.
- General search across the library

### Prospective

### Study Design

**Protocol Authoring**
Support the authoring of new protocols including complex studies such as Master, Umbrella etc.

This would include analyzing previous protocol information to inform design choices (see Protocol Store).

Writing associated SAP, Informed Consent, Monitoring Plan, Drug Plan etc via content reuse.

**Study Build**
Build of the study based on the protocol I including all detailed procedures and assessments to allow for immediate deployment.

**Feasibility & Cost**
Support study feasibility and complexity evaluations.

Determine the cost of the study including such items as vendor and CRO costs and support the laboratory RFP process

**Regulatory Authority**
Approval for the execution of the study including IND, CTA etc and discussions with regulatory authorities

**Stakeholder Views**
Tailored views for stakeholders such as IRB and ethic committees.

### Study Start-up

**Deployment**
Deployment of the study across the enterprise using the USDM to support electronic configuration rather than manual data entry including Data collection systems (incl. EDC), CTMS, IRT, Data transmission specifications and more

**TMF**
Linking the protocol to the specification of milestones and essential documents

**Data Collection Strategy**
Ensure all data that is needed for analysis is collected. Remove unnecessary procedures & data collection to reduce trial cost and patient burden.

**Trial Registries**
Registration of the study with registries such as CT.gov, CTIS etc.

**Training**
Provision of training materials to those involved in the study including sites

**Site Support**
Provision of tailored information to sites regarding study execution.

### Study Execution

**Amendments**
Support the amendment process with the ability to indicate the precise changes to the user community.

Ensure fewer amendments (see Protocol Store).

**Subject Information**
Provision of improved information to subjects regarding study execution, improve expectation management.

**Drug & Study Materials**
Assist in the planning and provision of study drug, materials and supplies

**Subject Recruitment**
Support the identification of subjects to help improve subject recruitment to the study. Help support EHR participation identification.

### Analysis & Reporting

**Automation**
Provide a solid foundation for the automation of analysis and reporting outputs such as TLFs and CSRs.

**SDTM**
Creation of 'T' domains. Auto generation of data domains

**ADaM**
Use the protocol statistical metadata to assist in the derivation of analysis datasets

**Statistical Analysis Plan**
Use the protocol as the initial source of the SAP to ensure consistency with objectives, endpoints, estimands

### Regulatory Submission

**Submission**
Preparation of the submission by the sponsor

**Regulatory Review**
Review of the submission by the regulatory authority and discussions / questions with the sponsor

**Statistical and Safety Review**
Support the comparison of data by arm or study design.

Allow for the verification of planned (protocol) data versus actual (study data) captured data

### Across The Lifecycle

**Dashboards**
The display of protocol related information sourced from one or more protocols in ways which have not been achievable before

**Comparison**
Compare and verify differences across versions and similar protocols

## Unified Study Definitions Model (USDM)

- Controlled Terms
- Unstructured Content
- Populations, Inclusion & Exclusion, Interventions & Indications, Estimands, Objectives & Endpoints
- Study, Identifiers, Amendments
- Study Designs, Arms, Epochs
- Procedures, Biomedical Concepts
- Detailed Study Logic, Encounters

**Study Details**
The overall study, its various versions, identifiers and associated governance. Also includes the amendments made to the study

**Study High Level Design**
The single or set of study designs making up the study detailing the epochs, arms etc.

**Study Science**
The detailed description of the study science: the populations and the associated inclusion and exclusion criteria, the indications being studied, the interventions being used and the objectives, endpoints and the associated estimands.

**Detailed Study Logic**
A precise definition of the study logic including support for the Schedule of Activities.

**Unstructured Content**
The ability to support one or more document presentations of the USDM content including the ICH M11 protocol template, sponsor templates and other documents.

**Procedures and Biomedical Concepts**
The detail around the procedures and observations to be performed as part of the detailed study designs.

**Controlled Terms**
The controlled terminology needed to define the semantics within the model. Managed in the same manner as all CDISC CT and aligned with the M11 template standard.

**NOTE:** The use cases presented are illustrative and the list is not intended to be exhaustive.

Infographic

Version 5, 24th October 2024. Prepared by D Iberson-Hurst for the TransCelerate 'DDF in Action' day.
With thanks to Rob Ferendo (TransCelerate), Bill Illis (Novartos), Jasmine Kestemont (Argenx), Kirsten Langendorf (d4k), Mary Lynn Mercado (Novartis), Lissa Morgan (Amgen), Johannes Ullander (d4k) and Peter Van Reusel (CDISC)

# TransCelerate Use Cases

## Protocol Store

- The Study Definition Repository

## Study Concept

- Optimization Scores
- Enrollment Forecasting
- Resourcing and Cost Predictions
- Country and Site Selection
- Timeline Forecasting
- Design Study
- Author Protocol
- Render Stakeholder View

## Study Start-Up

- Study Build
- Central Analysis Services
- Clinical Trial Materials
- Interactive Response Technology
- Clinical Trial Management
- Draft Study Budget
- Draft Site-Level Budgets

## Study Execution

- Report Protocol Changes
- Estimate Trial Site Inventory Needs
- Query Patient Screening Facilities
- Populate Trial Site Data Systems
- Verify Site Compliance

## Analysis & Reporting

- SDTM Trial Domains
- SDTM Data Domains
- SDTM to ADaM
- Perform Study Analysis
- Draft Statistical Analysis Plan

## Regulatory Submission

- Transmit Study Protocol
- Review Study Protocol
- Compare Actual to Planned Treatment Progression

Use Cases

# Summary

# Summary

FROM
**Document-First**
Silos • Manual handoffs

→

TO
**Data-First**
Single source • Automation

## STANDARDS CONVERGING

**ICH M11**
Template & Structure

**USDM v4**
Data Model & API

**HL7 FHIR**
Interoperability

## THREE PRIORITIES DRIVING ADOPTION

**1**
**CAPACITY MULTIPLIER**
Same resources, more studies. Automation eliminates re-keying—time saved is capacity gained.

**2**
**ENDLESS USE CASES**
EDC, regulatory, sites, feasibility, amendments, reuse... too many to list. Every downstream system benefits.

**3**
**AI-READY FOUNDATION**
Structured data + rich context = AI that works..

## ⚡ THE SCHEDULE OF ACTIVITIES IS THE ENGINE ⚡
A digital SoA isn't just a table—it's executable study logic. Timepoints, activities, conditions & relationships that power everything downstream.

### ⚠ MINDSET SHIFT REQUIRED

✗ Single source of truth
✗ Don't fit new tech into old processes
✗ Protocol approval as endpoint → continuum
✗ Mapping to old workflows won't unlock value

### ✓ SUCCESS FACTORS

✓ Maximize digital SoA
✓ Document to data paradigm
✓ Not quick wins, we want continuous wins, this will take time
✓ Change management, what is in it for me

d4k

# Useful Links

# Resources & Contact

**CDISC DDF Page**

Main CDISC web page

CDISC DDF

**CDISC GitHub**

The CDISC GitHub containing the USDM deliverables

CDISC

**Transcelerate DDF**

Main Transcelerate DDF webpage

TCB DDF

**Transcelerate DDF Pages**

Specific DDF web pages

TCB DDF

**Transcelerate SDR GitHub**

SDR GitHub

SDR GitHub

**d4k Useful Resources**

Set of useful USDM resources

Resources

LinkedIn